

V4.2 Toy Model 4D

Source-to-Curvature Calibration and Einstein-Tensor Consistency

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 4C

Status: Toy Model 4D tests whether the Einstein-tensor candidate G_{tt} from 4C can be calibrated against candidate relational source functions. It also checks the Bianchi/conservation condition numerically. It does not derive Newton's constant, physical stress-energy, Einstein Field Equations, gravity, or the dark sector.

1. Purpose

Toy Model 4C computed curvature from a controlled varying Lorentzian metric. Toy Model 4D asks whether the geometry side and a candidate source side can be consistently related.

The narrow target is not to prove Einstein's Field Equation. The target is to test candidate source-to-curvature calibrations and identify whether raw relational density is enough.

2. Critical correction before calibration

A scalar density profile $u(x,y)$ is not automatically the physical source in General Relativity. In GR the source is the stress-energy tensor $T_{\mu\nu}$, not simply a scalar bump. Also, curvature depends on derivatives of the metric, so G_{tt} is generally sensitive to gradients and second derivatives of the metric field, not only to the value of u itself.

Therefore, if G_{tt} is not strictly proportional to raw u , that is not a failure of the tensor machinery. It means the source model is too crude.

3. Controlled source and geometry

The same controlled metric from 4C was used:

$$u(x,y) = \text{epsilon} \exp[-(x^2 + y^2)/(2 \text{rho}^2)]$$

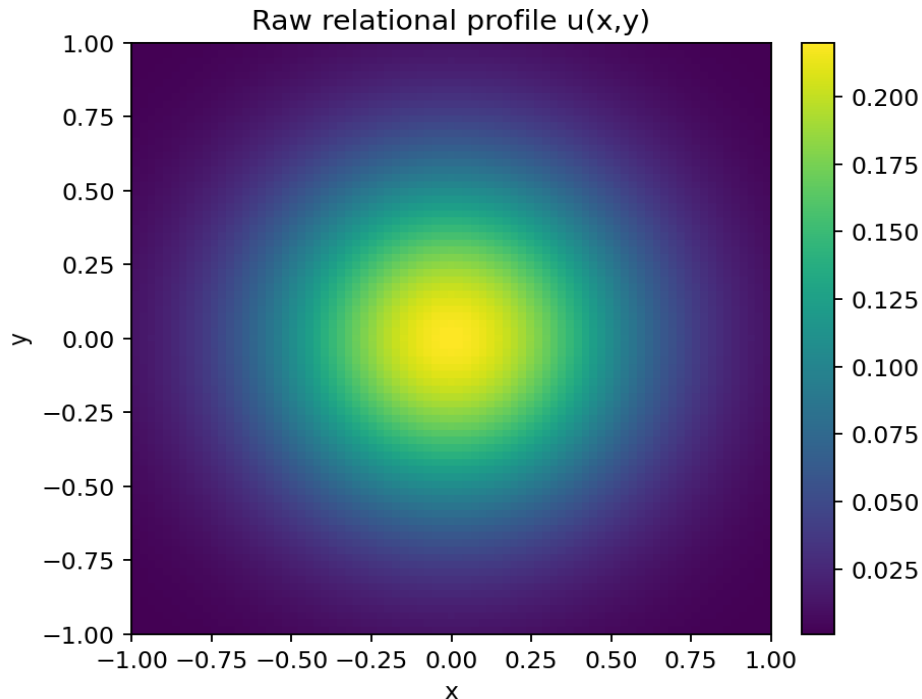
$$g_{tt} = -c^2(1 - 2u)$$

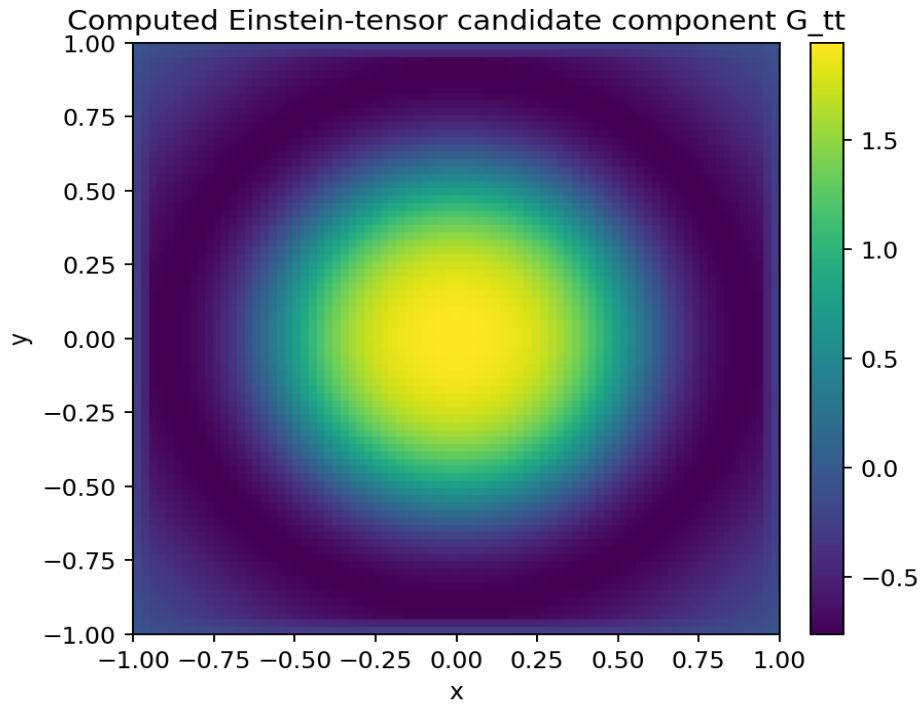
$$g_{xx} = 1 + 2u$$

$$g_{yy} = 1 + 2u$$

$$G_{\mu\nu} = R_{\mu\nu} - 1/2 g_{\mu\nu} R$$

The test compares the computed G_{tt} against several candidate source functions.





4. Proportionality test

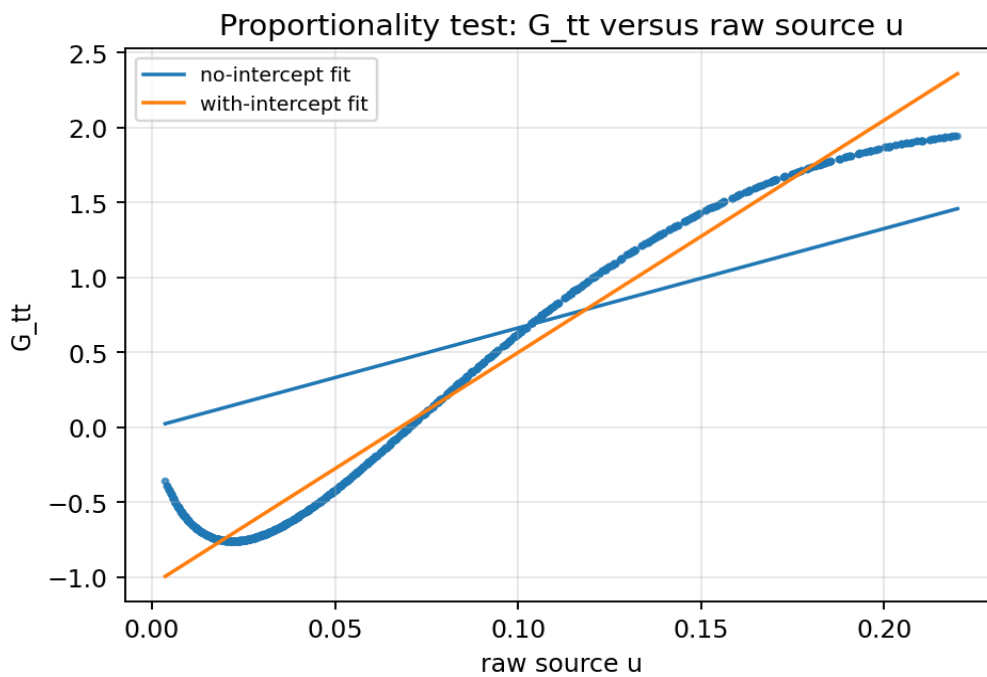
The basic calibration rule tested is:

$$G_{tt} \approx \kappa S$$

Meaning of symbols:

- G_{tt} is the time-time component of the Einstein-tensor candidate.
- S is the candidate relational source.
- κ is the fitted toy coupling constant.
- This is only a toy calibration. It is not Newton's gravitational constant and not the physical GR coupling.

First candidate: raw source $S = u$.



Raw-source calibration	Value
No-intercept κ for $G_{tt} \approx \kappa u$	6.628787e+00
No-intercept residual ratio	7.212144e-01
Correlation G_{tt} vs u	0.983875

Raw-source calibration	Value
With-intercept slope	1.549894e+01
With-intercept offset	-1.050933e+00
With-intercept residual ratio	1.772897e-01

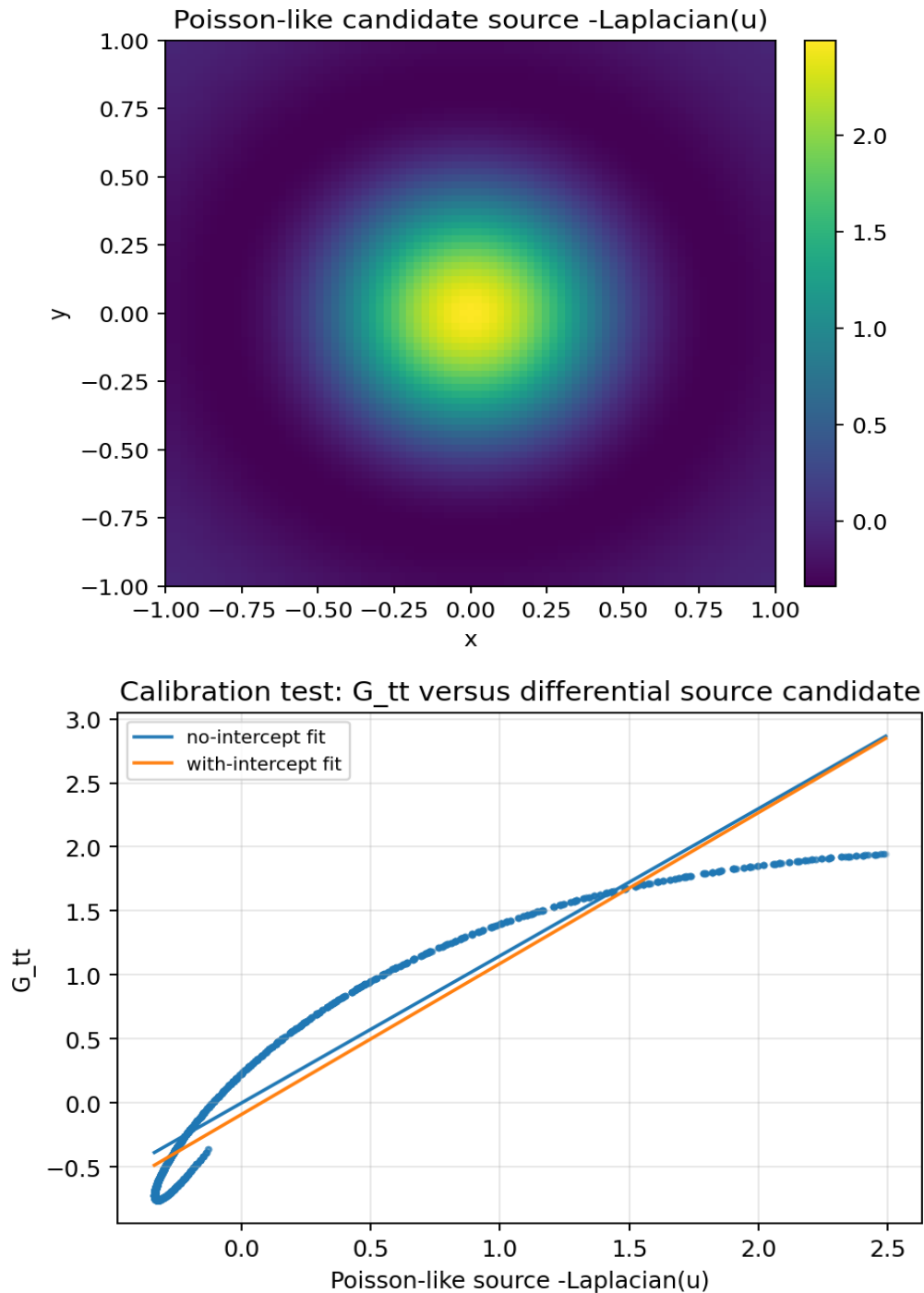
Result: raw u is correlated with G_tt, but strict direct proportionality is poor. The no-intercept residual is too large to treat u alone as the calibrated source.

5. Differential source candidate

Because curvature depends on metric derivatives, a Poisson-like candidate was also tested:

$$S_P = -\nabla^2 u$$

Meaning: $\nabla^2 u$ is the Laplacian, a measure of how much the source profile bends spatially. The minus sign makes the center of the Gaussian positive in this test.



Differential-source calibration	Value
No-intercept kappa for $G_{tt} \approx \kappa(-\nabla^2 u)$	1.149549e+00
No-intercept residual ratio	3.415403e-01

Differential-source calibration	Value
Correlation G_tt vs -lap u	0.943753
With-intercept slope	1.178571e+00
With-intercept offset	-8.957044e-02
With-intercept residual ratio	3.277520e-01

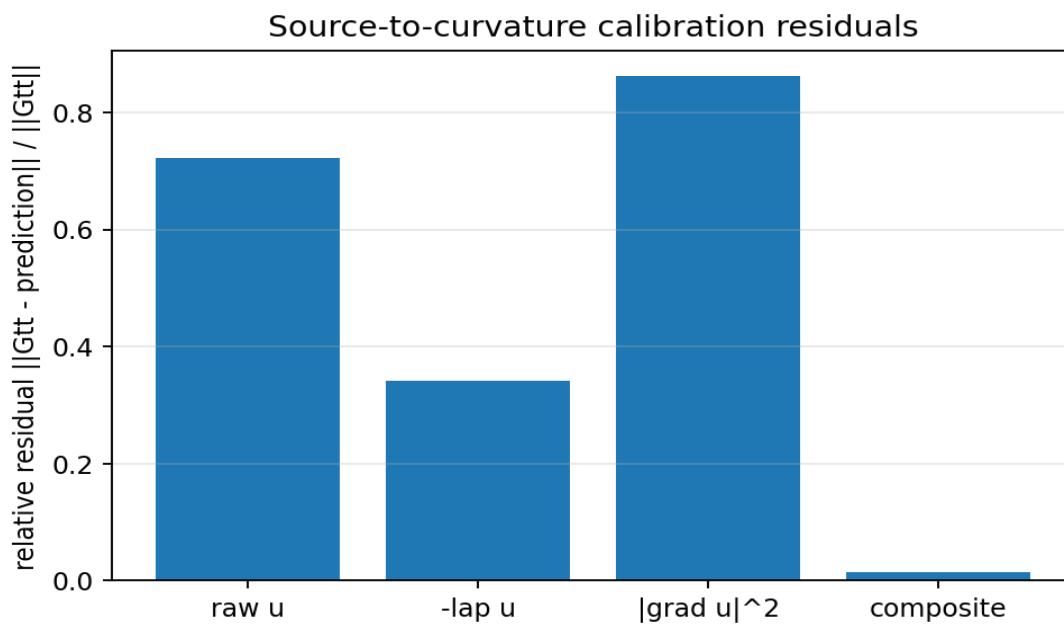
Result: the differential source is a better strict proportional source than raw u, but it is still not exact. This points toward a tensorial/differential source model rather than a raw scalar-density model.

6. Composite calibration

A small composite source basis was tested:

$$S_{\text{comp}} = a u + b(-\nabla^2 u) + c |\text{grad } u|^2 + d u^2$$

This is not proposed as a physical law. It is a diagnostic to test whether the curvature response contains both value and derivative information from the relational profile.



Composite coefficient	Value
a for u	3.899463e+01
b for -lap u	4.110785e+00
c for $ \text{grad } u ^2$	-8.751197e+00
d for u^2	-3.495691e+02
Composite residual ratio	1.413074e-02
Composite prediction correlation	0.999899

Interpretation: source-to-curvature calibration improves substantially when derivative and nonlinear terms are allowed. This confirms the source cannot be treated as raw density only. It does not identify a unique physical stress-energy tensor.

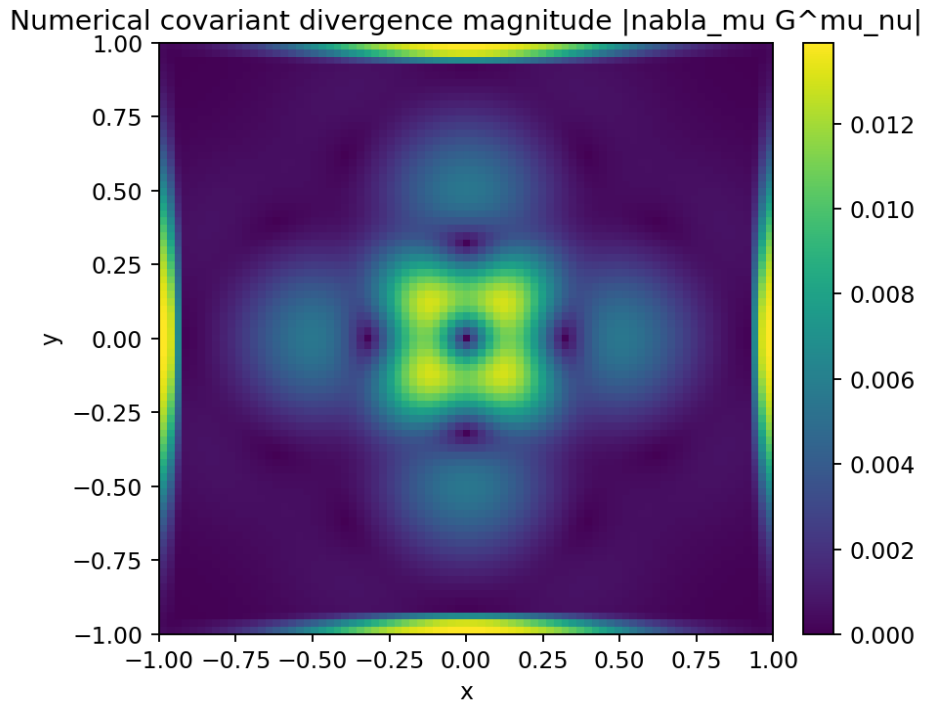
7. Bianchi / conservation check

For a true Einstein tensor constructed from a metric, the contracted Bianchi identity states:

$$\nabla_{\mu} G^{\mu}_{\nu} = 0$$

Plain meaning: the geometry side is automatically covariantly conserved. If one later sets $G_{\mu\nu} = \kappa T_{\mu\nu}$, then the source tensor $T_{\mu\nu}$ must also be covariantly conserved.

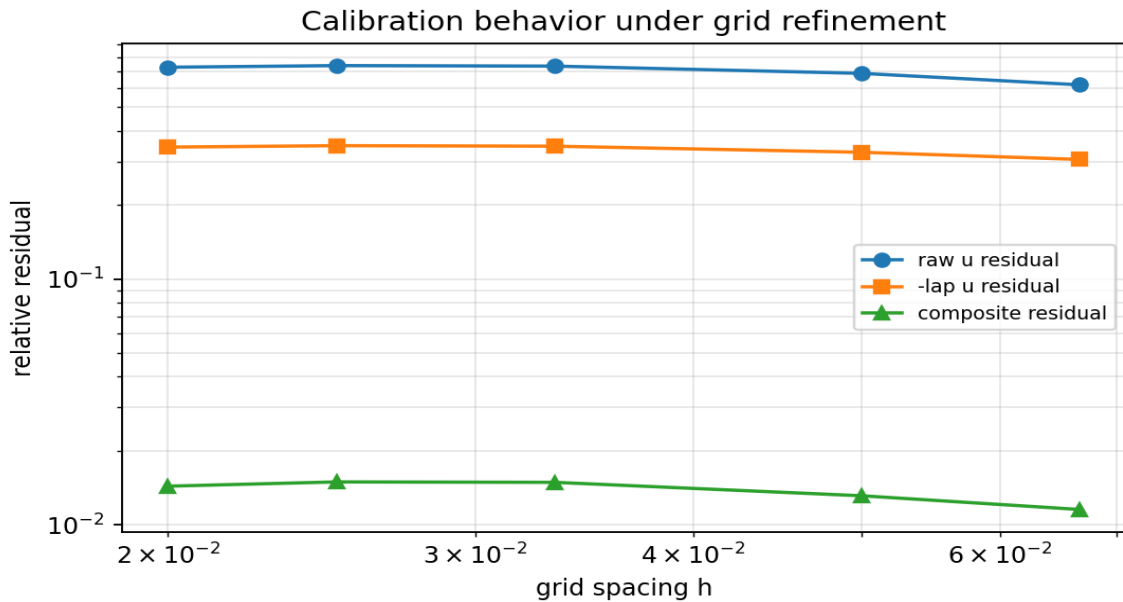
Toy Model 4D computes the numerical divergence of the mixed Einstein tensor G^{μ}_{ν} . Because this is a finite-difference grid, the result is not exactly zero, but the residual is small and behaves as a numerical consistency diagnostic.



Bianchi diagnostic	Value
Max $ \nabla_\mu G^\mu_\nu $ interior	1.299728e-02
Mean $ \nabla_\mu G^\mu_\nu $ interior	2.481676e-03
Relative max divergence diagnostic	1.304887e-04
Max component $\nu=t$	0.000000e+00
Max component $\nu=x$	1.089284e-02
Max component $\nu=y$	1.089284e-02

8. Grid-refinement behavior

The calibration residuals and conservation residual were checked across grid refinements.



N	h	raw u res.	-lap u res.	composite res.	max divG	Lorentz sig.
31	0.06667	6.165069e-01	3.065082e-01	1.153579e-02	8.156693e-02	529/529
41	0.05000	6.864015e-01	3.277121e-01	1.311735e-02	4.751799e-02	1089/1089
61	0.03333	7.347106e-01	3.470092e-01	1.487967e-02	2.256485e-02	2809/2809
81	0.02500	7.381145e-01	3.486676e-01	1.493918e-02	1.299728e-02	5041/5041
101	0.02000	7.269659e-01	3.439848e-01	1.435368e-02	8.389746e-03	7569/7569

9. Signature and core numerical record

Diagnostic	Value
Grid	81 x 81
Interior calibration points	4761
Lorentzian signatures	4761/4761
G _{tt} at center	1.945753e+00
u at center	2.200000e-01
-lap(u) at center	2.492123e+00
G _{tt} min interior	-7.598544e-01
G _{tt} max interior	1.945753e+00

10. Interpretation inside the V4.2 framework

Toy Model 4D supports this limited ladder:

```
4C metric curvature machinery
-> Einstein-tensor candidate Gmu_nu
-> candidate source functions S
-> calibration tests Gtt ~= $\kappa$  S
-> raw density alone fails strict proportionality
-> differential/composite source performs better
-> conservation condition checked through  $\nabla_\mu G^{\mu\nu}$ 
```

It does not yet support this stronger ladder:

```
relational density u
-> unique physical stress-energy tensor Tmu_nu
-> Einstein Field Equations
-> Newton's G
-> real gravity / dark sector explanation
```

11. What Toy Model 4D achieves

- It tests direct proportionality between G_{tt} and raw relational profile u.
- It shows raw u alone is not sufficient as a strict proportional source.
- It tests a Poisson-like differential source -laplacian(u), which performs better.
- It shows composite value/derivative features can fit the geometry response more closely.
- It computes the numerical covariant divergence of the Einstein-tensor candidate.
- It preserves Lorentzian signature across the tested calibration region.
- It identifies the next missing object: a proper relational stress-energy tensor candidate.

12. What Toy Model 4D does not achieve

- It does not derive Einstein Field Equations.
- It does not derive Newton's gravitational constant G.
- It does not derive the physical coupling constant κ .
- It does not derive physical stress-energy T_{mu_nu}.
- It does not prove that relational density physically causes curvature.
- It does not solve gravity.
- It does not explain dark matter or dark energy.
- It does not prove physical 3T.

13. Close-out statement for 4D

Toy Model 4D is a useful calibration test, but it does not close the gravity loop. The computed Einstein-tensor component G_{tt} is strongly related to the localized profile, but not strictly proportional to raw u. A derivative source such as -laplacian(u) performs better, and a composite differential source performs better still. The numerical Bianchi/conservation check is consistent at finite-difference precision. The correct conclusion is that relational source modeling must become tensorial and differential; raw density alone is too crude. The next step must define a candidate relational stress-energy tensor, not merely a scalar source bump.

14. Recommended next chapter

The next logical chapter is Toy Model 4E: Relational Stress-Energy Tensor Candidate.

1. Define source variables beyond raw density: density, flow, pressure/tension, anisotropy, and update rate.
2. Construct a toy $T_{\mu\nu}$ candidate from relational-network quantities.
3. Check covariant conservation: $\nabla_{\mu} T^{\mu\nu} \approx 0$.
4. Compare $G_{\mu\nu}$ to $\kappa T_{\mu\nu}$ component-by-component.
5. Only after a tensorial source candidate exists should a field-equation analogue be discussed.

15. Rebind prompt for continuation

We are resuming after Toy Model 4D.

4C computed curvature from a controlled varying Lorentzian metric:

```
u(x,y) = epsilon exp[-(x^2+y^2)/(2rho^2)]  
g_tt = -c^2(1 - 2u)  
g_xx = g_yy = 1 + 2u
```

4D tested source-to-curvature calibration.

Basic test:

$G_{tt} \approx \kappa S$

Candidate sources tested:

```
S_raw = u  
S_poisson = -laplacian(u)  
S_comp = a u + b(-laplacian u) + c|grad u|^2 + d u^2
```

Result:

Raw u is correlated with G_{tt} but fails strict no-intercept proportionality.

$-\text{laplacian}(u)$ performs better.

Composite differential source performs better still.

Bianchi/conservation check:

$\nabla_{\mu} G^{\mu\nu}$ was computed numerically and is small at finite-difference precision.

Critical guardrail:

4D does not derive Einstein Field Equations, physical stress-energy, Newton's G, gravity, dark matter, dark energy, or physical 3T.

Main lesson:

The source cannot be only a scalar relational-density bump.

The next missing object is a tensorial relational stress-energy candidate.

Next recommended chapter:

Toy Model 4E - Relational Stress-Energy Tensor Candidate.